

Project1_Final_Submission

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1 STAT 5243 Project 1: WallStreetBets Data Pipeline and Exploratory Analytics

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1.1 1. Introduction and Dataset Description

This project analyzes 53,187 posts from `r/wallstreetbets` during the 2020-2021 meme-stock period. The data combines structured engagement fields (`score`, `comms_num`, timestamps, URL type) with unstructured text (`title`, `body`), creating a mixed-modality dataset with meaningful cleaning and feature-engineering demands.

The final cleaned analytical table retains 53,187 rows and 30 columns after deterministic preprocessing policies.

1.2 2. Data Acquisition Methodology

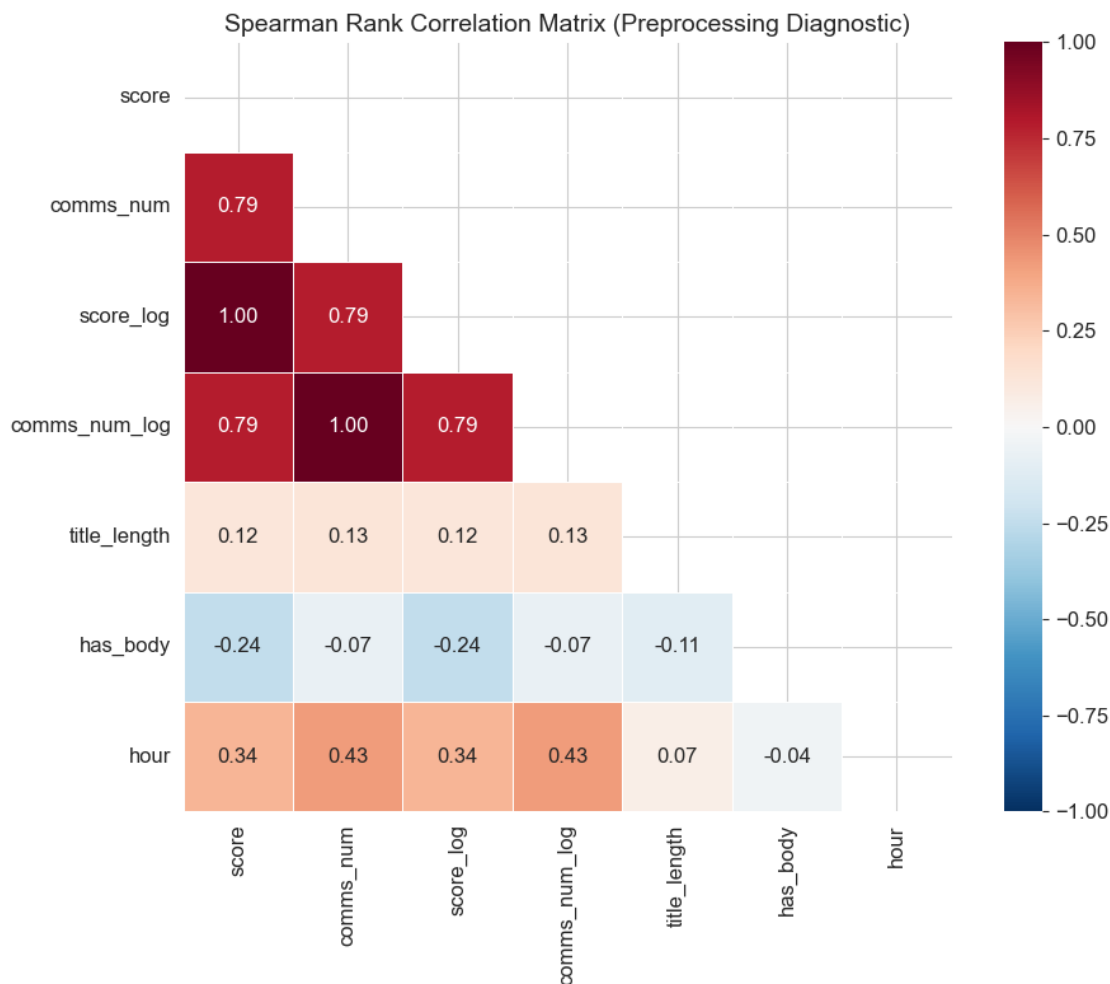
The source dataset was obtained from Kaggle (`[reddit_wsb.csv]`) and originally collected from Reddit `r/wallstreetbets` using PRAW (Python Reddit API Wrapper). We preserved the raw CSV unchanged in deliverables and produced a cleaned/engineered dataset via documented scripts and notebooks.

Acquisition artifact lineage is recorded in section JSON manifests under `Project Workspace/Supporting Materials/Report Sources/artifacts/json/`.

1.3 3. Cleaning and Preprocessing Steps

Cleaning policy highlights: (1) strict datetime coercion (`invalid_timestamp_count_in_raw = 0`), (2) structural missingness handling (`missing_body_pct = 53.49%`) via explicit indicators, (3) duplicate audit (`duplicate id/full-row count = 0`), (4) outlier-aware log transforms, and (5) dual scaling + categorical encoding for model readiness.

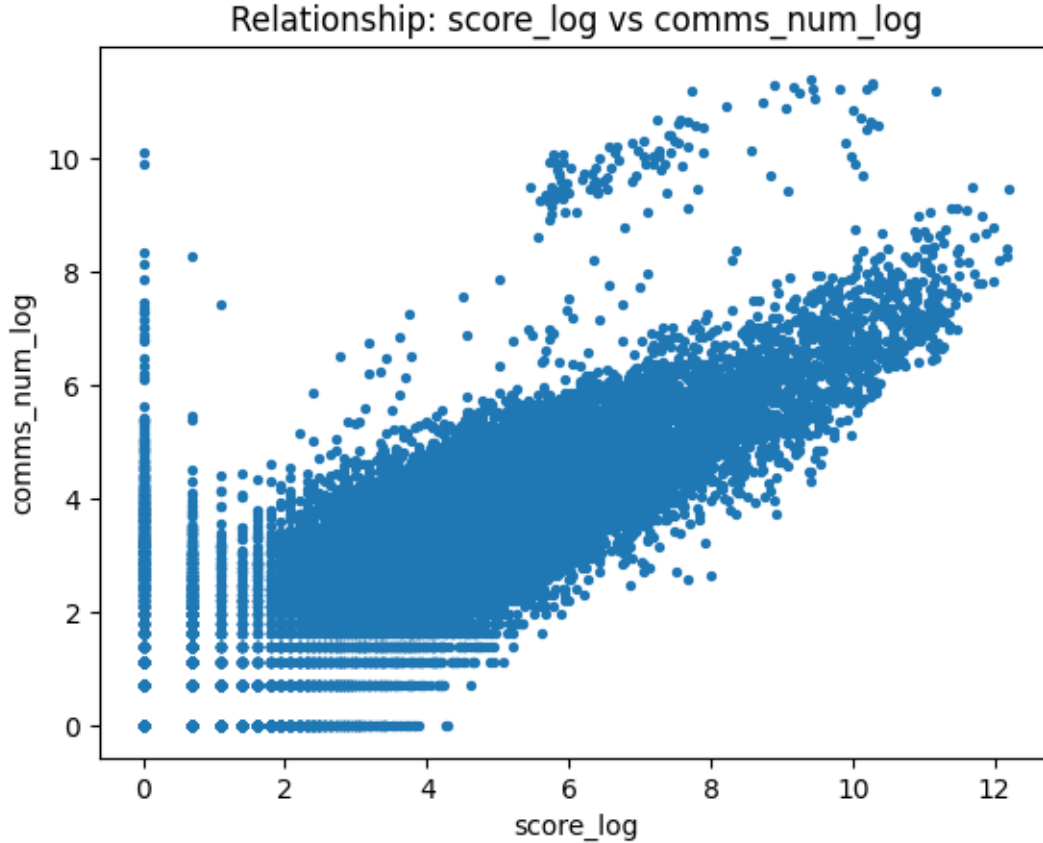
Evidence: - Figure: `.../artifacts/figures/02_cleaning_fig_01_missingness_diagnostics.png`
- JSON: `.../artifacts/json/02_cleaning_policy_metrics.json` -
Section details: `Project Workspace/Supporting Materials/Report Sources/sections/03_cleaning_preprocessing.tex`



1.4 4. Exploratory Data Analysis (EDA)

Advanced EDA quantified engagement dynamics beyond visual summaries. Key results: Spearman correlation between vote and comment activity ($\rho = 0.7852$, $p \ll 0.001$), post-type group difference significance (Kruskal-Wallis statistic 5855.658, $p \ll 0.001$), and viral-tail anomaly rate ($|z| \geq 3$ outlier rate 0.173%).

Interpretation: investor attention is clustered and format-sensitive, with rare but influential extreme events.

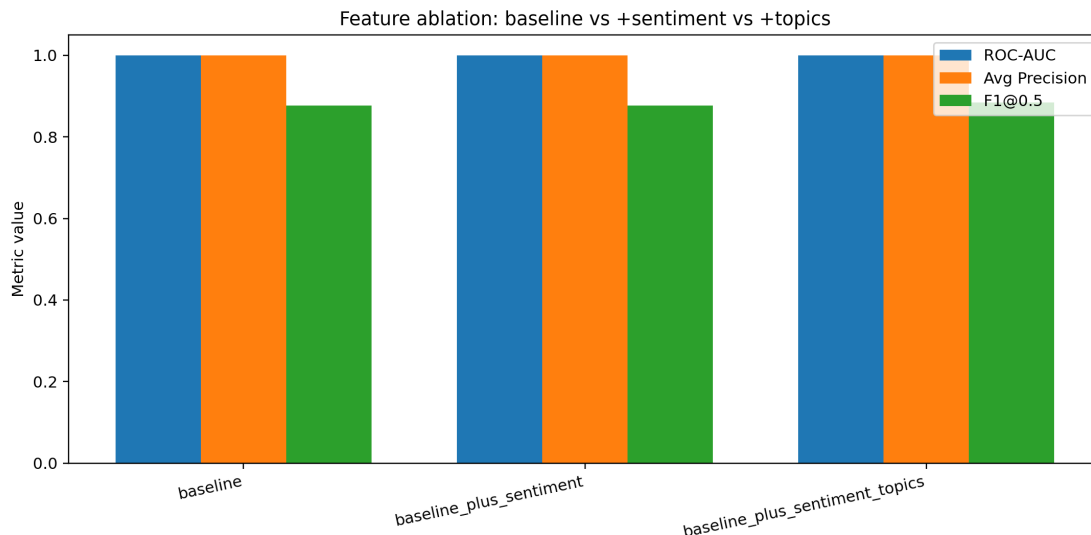


1.5 5. Feature Engineering Process and Justification

Feature engineering added lexicon sentiment and LDA topic probabilities to baseline engagement/temporal predictors, then evaluated deterministic ablation (seed 42).

Model	ROC-AUC	Avg Precision	F1@0.5
Baseline	0.999984	0.999710	0.876950
Baseline + Sentiment	0.999984	0.999710	0.876950
Baseline + Sentiment + Topics	0.999975	0.999546	0.884615

Measured lift: topic features improve thresholded F1 by +0.007666 absolute.



1.6 6. Summary of Key Findings

Cleaning policies preserved sample size while resolving inconsistencies and structural missingness. EDA showed strong engagement coupling and significant content-format effects. Feature engineering improved actionable classification behavior at threshold level, especially when topic features were added.

Claim-to-artifact traceability is documented in Project Workspace/Supporting Materials/Report Sources/main_report.tex and Project Workspace/qa/evidence_registry_report_depl

1.7 7. Challenges Faced and Future Recommendations

Challenges included high missingness in free-text body fields, heavy-tailed engagement distributions, and possible leakage-like ease because virality target is derived from score. Future work should test harder targets, add time-aware splits, and integrate richer NLP (contextual embeddings and sarcasm-aware sentiment).

1.8 8. GitHub Repository Link

- <https://github.com/ZemingLiang/STAT-5243-Project-1-Team-2>

1.9 9. Each Member's Contribution

- **Zeming Liang:** Data cleaning and preprocessing; creation/normalization of reddit_wsb.csv, reddit_wsb_cleaned.csv, and Cleaning-and-Preprocessing.ipynb; final repo organization and PNG/JSON/output integration.
- **Zuer Weng:** EDA analysis, advanced statistical diagnostics, and EDA branch artifacts.
- **Duoli Chen:** Report-writing narrative sections (introduction, acquisition, summary, challenges) and narrative refinement.
- **Isaac Beers:** Feature engineering workflow, viral-classification diagnostics, and model interpretation components.